We begin by defining the concept of Riemann integrability of real-valued functions defined on a closed bounded interval of $\mathbb{R}$, using the Riemann sums familiar to the reader from calculus. This method has the advantage that it extends immediately to the case of functions whose values are complex numbers, or vectors in the space $\mathbb{R}^n$. In Section 7.2, we will establish the Riemann integrability of several important classes of functions: step functions, continuous functions, and monotone functions. However, we will also see that there are functions that are *not* Riemann integrable. The Fundamental Theorem of Calculus is the principal result in Section 7.3. We will present it in a form that is slightly more general than is customary and does not require the function to be a derivative at every point of the interval. A number of important consequences of the Fundamental Theorem are also given. In Section 7.3 we also give a statement of the definitive Lebesgue Criterion for Riemann integrability. This famous result is usually not given in books at this level, since its proof (given in Appendix C) is somewhat complicated. However, its statement is well within the reach of students, who will also comprehend the power of this result. In Section 7.4, we discuss an alternative approach to the Riemann integral due to Gaston Darboux that uses the concepts of upper integral and lower integral. The two approaches appear to be quite different, but in fact they are shown to be equivalent. The final section presents several methods of approximating integrals, a subject that has become increasingly important during this era of high-speed computers. While the proofs of these results are not particularly difficult, we defer them to Appendix D.

An interesting history of integration theory, including a chapter on the Riemann integral, is given in the book by Hawkins cited in the References.

Section 7.1 Riemann Integral

We will follow the procedure commonly used in calculus courses and define the Riemann integral as a kind of limit of the Riemann sums as the norm of the partitions tend to 0. Since we assume that the reader is familiar—at least informally—with the integral from a calculus course, we will not provide a motivation of the integral, or discuss its interpretation as the "area under the graph," or its many applications to physics, engineering, economics, etc. Instead, we will focus on the purely mathematical aspects of the integral.

However, we first define some basic terms that will be frequently used.

Partitions and Tagged Partitions

If $I := [a, b]$ is a closed bounded interval in $\mathbb{R}$, then a **partition** of I is a finite, ordered set $\mathcal{P} := (x_0, x_1, \ldots, x_{n-1}, x_n)$ of points in I such that

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

(See Figure 7.1.1.) The points of $\mathcal{P}$ are used to divide $I = [a, b]$ into non-overlapping subintervals

$$I_1 := [x_0, x_1], \quad I_2 := [x_1, x_2], \ldots, \quad I_n := [x_{n-1}, x_n].$$

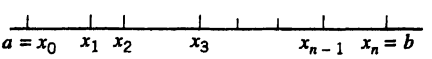


Figure 7.1.1 A partition of $[a, b]$

Often we will denote the partition $\mathcal{P}$ by the notation $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$. We define the **norm** (or **mesh**) of $\mathcal{P}$ to be the number

$$(1) \qquad \|\mathcal{P}\| := \max\{x_1 - x_0, x_2 - x_1, \ldots, x_n - x_{n-1}\}.$$

Thus the norm of a partition is merely the length of the largest subinterval into which the partition divides $[a, b]$. Clearly, many partitions have the same norm, so the partition is *not* a function of the norm.

If a point t_i has been selected from each subinterval $I_i = [x_{i-1}, x_i]$, for $i = 1, 2, \ldots, n$, then the points are called **tags** of the subintervals I_i. A set of ordered pairs

$$\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$$

of subintervals and corresponding tags is called a **tagged partition** of I; see Figure 7.1.2. (The dot over the $\mathcal{P}$ indicates that a tag has been chosen for each subinterval.) The tags can be chosen in a wholly arbitrary fashion; for example, we can choose the tags to be the left endpoints, or the right endpoints, or the midpoints of the subintervals, etc. Note that an endpoint of a subinterval can be used as a tag for two consecutive subintervals. Since each tag can be chosen in infinitely many ways, each partition can be tagged in infinitely many ways. The norm of a tagged partition is defined as for an ordinary partition and does not depend on the choice of tags.

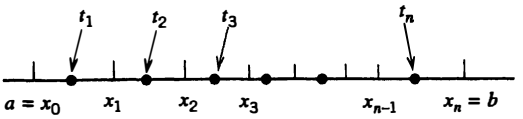


Figure 7.1.2 A tagged partition of $[a, b]$

If $\dot{\mathcal{P}}$ is the tagged partition given above, we define the **Riemann sum** of a function $f : [a, b] \to \mathbb{R}$ corresponding to $\dot{\mathcal{P}}$ to be the number

$$(2) \qquad S(f; \dot{\mathcal{P}}) := \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}).$$

We will also use this notation when $\dot{\mathcal{P}}$ denotes a *subset* of a partition, and not the entire partition.

The reader will perceive that if the function f is positive on $[a, b]$, then the Riemann sum (2) is the sum of the areas of n rectangles whose bases are the subintervals $I_i = [x_{i-1}, x_i]$ and whose heights are $f(t_i)$. (See Figure 7.1.3.)

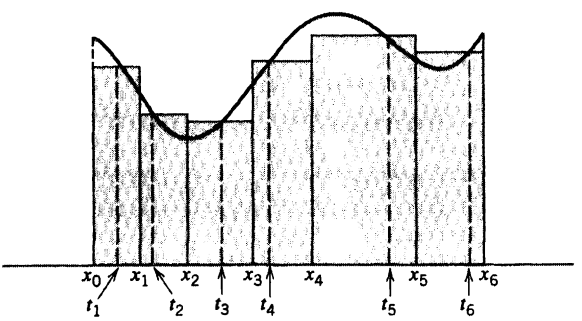


Figure 7.1.3 A Riemann sum

Definition of the Riemann Integral

We now define the Riemann integral of a function f on an interval $[a, b]$.

7.1.1 Definition A function $f : [a, b] \to \mathbb{R}$ is said to be **Riemann integrable** on $[a, b]$ if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition of $[a, b]$ with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$\left|S(f; \dot{\mathcal{P}}) - L\right| < \varepsilon.$$

The set of all Riemann integrable functions on $[a, b]$ will be denoted by $\mathcal{R}[a, b]$.

Remark It is sometimes said that the integral L is "the limit" of the Riemann sums $S(f : \dot{\mathcal{P}})$ as the norm $||\dot{\mathcal{P}}|| \to 0$. However, since $S(f; \dot{\mathcal{P}})$ is not a function of $||\dot{\mathcal{P}}||$, this **limit** is not of the type that we have studied before.

First we will show that if $f \in \mathcal{R}[a, b]$, then the number L is uniquely determined. It will be called the **Riemann integral of** f over $[a, b]$. Instead of L, we will usually write

$$L = \int_a^b f \quad \text{or} \quad \int_a^b f(x)dx.$$

It should be understood that any letter other than x can be used in the latter expression, so long as it does not cause any ambiguity.

7.1.2 Theorem *If $f \in \mathcal{R}[a, b]$, then the value of the integral is uniquely determined.*

Proof. Assume that L' and L'' both satisfy the definition and let $\varepsilon > 0$. Then there exists $\delta'_{\varepsilon/2} > 0$ such that if $\dot{\mathcal{P}}_1$ is any tagged partition with $||\dot{\mathcal{P}}_1|| < \delta'_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}_1) - L'\right| < \varepsilon/2.$$

Also there exists $\delta''_{\varepsilon/2} > 0$ such that if $\dot{\mathcal{P}}_2$ is any tagged partition with $||\dot{\mathcal{P}}_2|| < \delta''_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}_2) - L''\right| < \varepsilon/2.$$

Now let $\delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0$ and let $\dot{\mathcal{P}}$ be a tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$. Since both $||\dot{\mathcal{P}}|| < \delta'_{\varepsilon/2}$ and $||\dot{\mathcal{P}}|| < \delta''_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}) - L'\right| < \varepsilon/2 \qquad \text{and} \qquad \left|S(f; \dot{\mathcal{P}}) - L''\right| < \varepsilon/2,$$

whence it follows from the Triangle Inequality that

$$\begin{aligned} |L' - L''| &= \left|L' - S(f; \dot{\mathcal{P}}) + S(f; \dot{\mathcal{P}}) - L''\right| \\ &\leq \left|L' - S(f; \dot{\mathcal{P}})\right| + \left|S(f; \dot{\mathcal{P}}) - L''\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, it follows that $L' = L''$. Q.E.D.

The next result shows that changing a function at a finite number of points does not affect its integrability nor the value of its integral.

7.1.3 Theorem If g is Riemann integrable on $[a, b]$ and if $f(x) = g(x)$ except for a finite number of points in $[a, b]$, then f is Riemann integrable and $\int_a^b f = \int_a^b g$.

Proof. If we prove the assertion for the case of one exceptional point, then the extension to a finite number of points is done by a standard induction argument, which we leave to the reader.

Let c be a point in the interval and let $L = \int_a^b g$. Assume that $f(x) = g(x)$ for all $x \neq c$. For any tagged partition $\dot{\mathcal{P}}$, the terms in the two sums $S(f; \dot{\mathcal{P}})$ and $S(g; \dot{\mathcal{P}})$ are identical with the exception of at most two terms (in the case that $c = x_i = x_{i-1}$ is an endpoint). Therefore, we have

$$|S(f; \dot{\mathcal{P}}) - S(g; \dot{\mathcal{P}})| = |\Sigma(f(x_i) - g(x_i))(x_i - x_{i-1})| \leq 2(|g(c)| + |f(c)|)\,||\dot{\mathcal{P}}||.$$

Now, given $\varepsilon > 0$, we let $\delta_1 > 0$ satisfy $\delta_1 < \varepsilon/(4(|f(c)| + |g(c)|)$, and let $\delta_2 > 0$ be such that $||\dot{\mathcal{P}}|| < \delta_2$ implies $|S(g; \dot{\mathcal{P}}) - L| < \varepsilon/2$. We now let $\delta := \min\{\delta_1, \delta_2\}$. Then, if $||\dot{\mathcal{P}}|| < \delta$, we obtain

$$|S(f; \dot{\mathcal{P}}) - L| \leq |S(f; \dot{\mathcal{P}}) - S(g; \dot{\mathcal{P}})| + |S(g; \dot{\mathcal{P}}) - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence, the function f is integrable with integral L. Q.E.D.

Some Examples

If we use only the definition, in order to show that a function f is Riemann integrable we must (i) know (or guess correctly) the value L of the integral, and (ii) construct a δ_ε that will suffice for an arbitrary $\varepsilon > 0$. The determination of L is sometimes done by calculating Riemann sums and guessing what L must be. The determination of δ_ε is likely to be difficult.

In actual practice, we usually show that $f \in \mathcal{R}[a, b]$ by making use of some of the theorems that will be given later.

7.1.4 Examples **(a)** Every constant function on $[a, b]$ is in $\mathcal{R}[a, b]$.

Let $f(x) := k$ for all $x \in [a, b]$. If $\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is any tagged partition of $[a, b]$, then it is clear that

$$S(f; \dot{\mathcal{P}}) = \sum_{i=1}^{n} k(x_i - x_{i-1}) = k(b - a).$$

Hence, for any $\varepsilon > 0$, we can choose $\delta_\varepsilon := 1$ so that if $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$|S(f; \dot{\mathcal{P}}) - k(b - a)| = 0 < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that $f \in \mathcal{R}[a, b]$ and $\int_a^b f = k(b - a)$.

(b) Let $g : [0, 3] \to \mathbb{R}$ be defined by $g(x) := 2$ for $0 \leq x \leq 1$, and $g(x) := 3$ for $1 < x \leq 3$. A preliminary investigation, based on the graph of g (see Figure 7.1.4), suggests that we might expect that $\int_0^3 g = 8$.

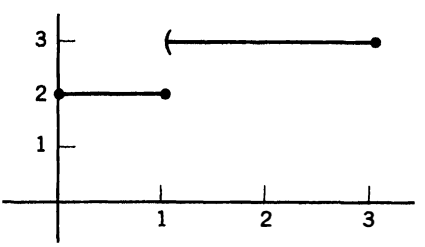


Figure 7.1.4 Graph of g